

PATENT SPECIFICATION

A Multi-Layer Transaction Processing System for Compliance Enforcement in Decentralised Finance Networks

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1. TECHNICAL FIELD

The present invention relates to the field of financial technology, and more particularly to systems and methods for enforcing anti-money laundering (AML) and regulatory compliance in decentralised finance transactions using blockchain smart contracts, machine learning risk scoring, and zero-knowledge cryptographic proofs.

2. BACKGROUND OF THE INVENTION

2.1. Technical Problem

Decentralised finance (DeFi) protocols process financial transactions on blockchain networks without centralised intermediaries. While this architecture offers advantages in transparency and accessibility, it creates significant regulatory compliance challenges. Current compliance approaches suffer from several deficiencies:

- (a) **Retroactive enforcement:** Existing blockchain analytics platforms (e.g., Chainalysis, Elliptic) operate as retrospective monitoring tools. They identify suspicious transactions *after* settlement, by which time value transfer has already occurred and may be irrecoverable.
- (b) **Platform-level rather than protocol-level compliance:** Conventional KYC/AML checks are performed at the exchange or platform level. When a user transacts through a DeFi protocol directly, these checks are bypassed entirely. Compliance is bound to the platform, not to the transaction.
- (c) **Reliance on labelled training data:** Supervised fraud detection models require large quantities of labelled fraudulent transactions. In practice, fewer than 0.1% of blockchain transactions are labelled as illicit, creating severe class imbalance and data scarcity problems.

- (d) **Privacy-compliance tension:** Compliance verification traditionally requires disclosure of sensitive personal and financial data. This conflicts with the pseudonymous nature of blockchain transactions and with data protection regulations such as the UK Data Protection Act 2018 and the EU General Data Protection Regulation.

2.2. Prior Art Limitations

Prior art systems known to the applicant include:

- Blockchain analytics platforms that provide post-hoc transaction tracing and address clustering but do not enforce compliance at the point of transaction.
- Exchange-level compliance systems that implement KYC/AML at centralised access points but cannot extend to protocol-level DeFi transactions.
- Standard machine learning fraud detection systems that require extensive labelled datasets and do not integrate with blockchain smart contract enforcement mechanisms.
- Zero-knowledge proof systems used in cryptocurrency privacy (e.g., Zcash) that conceal transaction details but do not implement compliance verification.

None of the prior art systems provides a unified architecture that combines pre-settlement compliance enforcement, weak supervision for label-scarce environments, graph-based analytics, sanctions screening, deterministic policy enforcement via smart contracts, and privacy-preserving verification via zero-knowledge proofs.

3. SUMMARY OF THE INVENTION

The present invention provides a multi-layer transaction processing system that enforces regulatory compliance *before* on-chain settlement in decentralised finance networks. The system addresses the limitations of the prior art through the following technical contributions:

- (i) A four-layer architecture comprising interface, compliance orchestration, offline training, and infrastructure layers that together implement a deterministic compliance lifecycle.
- (ii) A weak supervision machine learning pipeline comprising a composite teacher ensemble that generates pseudo-labels from unlabelled blockchain transaction data, enabling the training of a student model without requiring labelled fraud data.
- (iii) A deterministic compliance decision matrix that maps continuous risk scores to discrete policy actions (approve, review, escrow, block) via fixed thresholds, ensuring reproducible and auditable decisions.
- (iv) A zero-knowledge proof subsystem (zkNAF) comprising three proof circuits—sanctions non-membership, risk range, and KYC verification—that enable privacy-preserving compliance verification on-chain.

- (v) Smart contract infrastructure that enforces compliance decisions deterministically, including automatic escrow of high-risk transactions and blocking of transactions exceeding risk thresholds.
- (vi) A state-adaptive transaction flow control module configured to compute network instability metrics and to select, from a predetermined set of enforcement profiles, one or more enforcement parameters that modulate the deterministic compliance decision process in response to the computed system state.

4. DETAILED DESCRIPTION OF THE INVENTION

4.1. System Architecture Overview

The system comprises four hierarchically organised processing layers, described below from the outermost (user-facing) to the innermost (infrastructure):

Layer I — Interface Layer. This layer provides the entry points through which external systems interact with the compliance system. It comprises:

- Client software development kits (SDKs) in TypeScript and Python that expose methods for submitting transactions, querying risk scores, and verifying compliance status.
- A RESTful application programming interface (API) providing HTTP endpoints for transaction submission, risk score retrieval, compliance status queries, and administrative functions.
- A web-based dashboard application providing visual interfaces for compliance monitoring, transaction review, and system administration.

Layer II — Compliance Orchestration Layer. This layer implements the core compliance logic. It comprises:

- A machine learning risk scoring engine that computes a numerical risk score for each submitted transaction based on engineered features derived from address-level, transaction-level, and graph-level attributes.
- A graph analytics module that constructs and analyses transaction graphs using a graph database, computing graph-level features including degree centrality, betweenness centrality, clustering coefficient, and community membership.
- A sanctions screening module that checks transaction counterparties against maintained sanctions lists including OFAC (Office of Foreign Assets Control) and EU sanctions databases.
- A policy engine that applies a deterministic compliance decision matrix to map computed risk scores to policy actions.
- A state-adaptive transaction flow control module configured to compute one or more network instability metrics and to select a predetermined enforcement profile that determines one or more enforcement parameters used by the policy engine.

Layer III — Offline Training Layer. This layer implements the machine learning model training pipeline. It comprises:

- A composite teacher ensemble for generating pseudo-labels on unlabelled data.
- A student model training pipeline using the pseudo-labels.
- Cross-validation and calibration procedures.

Layer IV — Infrastructure Layer. This layer provides the persistent data storage, blockchain smart contracts, and cryptographic proof systems. It comprises:

- Blockchain smart contracts deployed on an Ethereum-compatible network that enforce compliance decisions deterministically.
- A zero-knowledge proof subsystem (zkNAF).
- Database systems for transaction records, graph data, and audit trails.
- Distributed file storage for immutable audit records.

4.2. Composite Teacher Weak Supervision Pipeline

The offline training layer implements a novel weak supervision approach to address the scarcity of labelled fraud data in blockchain transactions. The composite teacher ensemble generates pseudo-labels as follows.

Three independent teacher models are constructed:

- (a) An autoencoder–gradient boosting teacher ($s_{\text{AE-XGB}}$) that trains an autoencoder on transaction features to learn a compressed representation, then trains a gradient boosting model on the reconstruction errors and latent features to estimate anomaly scores.
- (b) A rule-based FATF teacher (s_{FATF}) that encodes Financial Action Task Force red-flag indicators as deterministic scoring rules, including rapid fund movement patterns, round-number transactions, structuring behaviour, and high-risk jurisdiction involvement.
- (c) A graph-based teacher (s_{Graph}) that computes suspicion scores based on graph-structural properties of the transaction network, including proximity to known high-risk addresses, unusual connectivity patterns, and temporal clustering of transactions.

The composite teacher score for each transaction x is computed as:

$$s_{\text{teacher}}(x) = w_1 \cdot s_{\text{AE-XGB}}(x) + w_2 \cdot s_{\text{FATF}}(x) + w_3 \cdot s_{\text{Graph}}(x) \quad (1)$$

where in the preferred embodiment, $w_1 = 0.4$, $w_2 = 0.3$, $w_3 = 0.3$.

The composite teacher scores are thresholded to generate binary pseudo-labels, which are then used to train a student model. In the preferred embodiment, the student model is an XGBoost gradient boosting classifier trained on 167 engineered features.

4.3. Risk Score Calibration

The raw output of the student model is recalibrated using a sigmoid recalibration function:

$$\hat{p}_{\text{cal}}(x) = \sigma(k \cdot (s(x) - P_{75})) \quad (2)$$

where σ is the logistic sigmoid function, k is a steepness parameter (in the preferred embodiment, $k = 30$), and P_{75} is the 75th percentile of the training score distribution. This recalibration compresses mid-range scores and amplifies boundary decisions, improving discrimination at the critical decision thresholds.

The calibrated probability is then mapped to an integer risk score on a 0–1000 scale for use by the compliance decision matrix.

4.4. Compliance Decision Matrix

The policy engine applies a deterministic decision matrix that maps integer risk scores to policy actions:

Score Range	Policy Action	Smart Contract Effect
[0, 200)	APPROVE	Transaction executed immediately
[200, 400)	APPROVE with monitoring	Transaction executed; monitoring flag set
[400, 600)	REVIEW	Transaction queued for manual review
[600, 800)	ESCROW	Funds locked in escrow smart contract
[800, 1000]	BLOCK	Transaction rejected on-chain

These thresholds are fixed and public, ensuring that compliance decisions are reproducible, auditable, and deterministic. Every transaction receives identical treatment for identical risk scores, regardless of the time, venue, or identity of the transacting parties.

4.5. State-Adaptive Transaction Flow Control (Adaptive Friction)

In an enhanced embodiment, the compliance orchestration layer comprises a state-adaptive transaction flow control module (also referred to herein as an adaptive friction controller). The purpose of this module is to modulate the enforcement strictness of the deterministic compliance lifecycle in response to system-wide instability, while preserving determinism and auditability.

The module computes one or more network instability metrics from:

- (a) a transaction graph constructed from recent on-chain activity; and/or
- (b) aggregate transaction-risk dynamics comprising one or more of: risk score distribution shift, high-risk transaction rate, graph centrality concentration, and temporal burst indicators.

Based on the computed instability metrics, the module selects an enforcement profile from a predetermined set of profiles (for example: *stable*, *stressed*, and *critical*). Each profile deterministically defines one or more enforcement parameters used by the policy engine and/or smart contracts, including:

- (a) one or more risk score thresholds used by the decision matrix;

- (b) a transaction rate limit or queue delay for review actions;
- (c) an escrow duration or release condition window; and
- (d) additional proof requirements (e.g., requiring a risk-range proof or sanctions proof for specific classes of transactions).

Accordingly, for a fixed system state (instability metrics and profile selection inputs), the selected enforcement profile and all resulting enforcement decisions are deterministic and reproducible.

4.6. Zero-Knowledge Proof Subsystem (zkNAF)

The zero-knowledge non-disclosing anti-fraud (zkNAF) subsystem enables participants to prove compliance properties without revealing underlying sensitive data. The subsystem implements three distinct proof circuits using the Groth16 zero-knowledge succinct non-interactive argument of knowledge (zk-SNARK) protocol:

4.6.1 Sanctions Non-Membership Proof

This circuit proves that a blockchain address is not a member of a set of sanctioned addresses, without revealing the address being verified. The circuit operates as follows:

1. The sanctioned address set is organised as a Merkle tree using the Poseidon hash function (chosen for its efficiency within arithmetic circuits).
2. The prover demonstrates possession of a valid Merkle non-membership proof—i.e., that the address does not appear at any leaf of the tree.
3. The verifier checks the proof against the published Merkle root of the current sanctions list without learning the prover's address.

4.6.2 Risk Range Proof

This circuit proves that a computed risk score falls within a specified acceptable range, without revealing the exact score. Formally, the prover demonstrates:

$$\text{score}_{\min} \leq s(x) \leq \text{score}_{\max} \quad (3)$$

where $s(x)$ is the computed risk score and $[\text{score}_{\min}, \text{score}_{\max}]$ is the acceptable range, without revealing $s(x)$.

4.6.3 KYC Verification Proof

This circuit proves that a participant has completed Know Your Customer verification with an accredited provider without revealing the participant's identity data. The circuit verifies a digital commitment to the KYC status and provider attestation.

4.6.4 Proof Lifecycle

Each zero-knowledge proof has a time-to-live of 24 hours, after which it must be regenerated. Proofs are verified on-chain by a dedicated verifier smart contract that stores verification results. The verification function accepts the proof components (π_a , π_b , π_c) and public inputs, performs the bilinear pairing check, and records the result with a timestamp.

4.7. Smart Contract Architecture

The infrastructure layer comprises a plurality of smart contracts deployed on an Ethereum-compatible blockchain network. The principal contracts include:

- **Core Contract:** Receives transaction risk scores from the compliance orchestration layer and registers compliance decisions on-chain. Implements the deterministic decision matrix as an on-chain mapping from score ranges to policy actions.
- **Escrow Contract:** Implements a time-locked escrow mechanism for transactions assigned the ESCROW policy action. Funds are held for a configurable review period during which authorised reviewers may release the funds (if the review outcome is favourable) or initiate a dispute resolution process.
- **ZK Verifier Contract:** Verifies Groth16 zero-knowledge proofs on-chain. Accepts proof components and public inputs, performs the elliptic-curve pairing verification, and stores the binary verification result with a timestamp for subsequent reference by other contracts.
- **Cross-Chain Bridge Contract:** Extends compliance enforcement to cross-chain transactions by implementing a bridge mechanism that requires compliance verification on both the source and destination chains before releasing bridged funds.
- **Dispute Resolution Contract:** Implements a structured process for resolving disputes arising from escrowed transactions, including escalation procedures and release conditions.

4.8. Feature Engineering

The machine learning risk scoring engine operates on 167 engineered features derived from three categories:

- (a) **Address-level features:** Transaction count, total value transferred, mean and variance of transaction amounts, age of address, number of unique counterparties, and temporal activity patterns.
- (b) **Transaction-level features:** Transaction value, gas parameters, input data characteristics, number of internal transactions, and token transfer indicators.
- (c) **Graph-level features:** Degree centrality, betweenness centrality, closeness centrality, clustering coefficient, PageRank score, community membership identifier, and local graph density.

4.9. Performance Characteristics

In the preferred embodiment, trained on 2.64 million Ethereum transactions, the system achieves the following performance:

Metric	Value
Meta-Ensemble ROC-AUC (test)	1.0000
Meta-Ensemble ROC-AUC (5-fold CV)	0.9965 ± 0.0049
Meta-Ensemble PR-AUC	0.9999
Transaction-Level XGBoost ROC-AUC	0.9878
Transaction-Level XGBoost F1	0.901
Cross-domain mean ROC-AUC (9 datasets)	0.9962
Escrow smart contract gas cost	287,890 gas

5. CLAIMS

The following claims define the scope of protection sought. Where a claim refers to another claim, the dependency is by claim number. Features of any dependent claim may be combined with any independent claim or with other dependent claims.

1. A computer-implemented transaction compliance system for decentralised finance networks, the system comprising:
 - (a) an interface layer configured to receive transaction requests from client applications via software development kits or application programming interfaces;
 - (b) a compliance orchestration layer comprising:
 - (i) a machine learning risk scoring engine configured to compute a numerical risk score for each received transaction request based on a plurality of engineered features derived from address-level, transaction-level, and graph-level attributes;
 - (ii) a graph analytics module configured to construct and analyse transaction graphs and compute graph-level features;
 - (iii) a sanctions screening module configured to check transaction counterparties against maintained sanctions databases; and
 - (iv) a state-adaptive transaction flow control module configured to compute one or more network instability metrics from recent transaction activity and to select, from a predetermined set of enforcement profiles, an enforcement configuration comprising:
 - (a) a risk threshold profile defining numerical thresholds for mapping risk scores to policy actions; and
 - (b) a proof-requirement policy defining, for at least one risk score tier or transaction class, one or more cryptographic proofs that are required to be verified as a prerequisite to settlement;
 - and
 - (v) a deterministic policy engine configured to:
 - (a) map computed risk scores to policy actions selected from the group consisting of: approve, approve with monitoring, review, escrow, and block, according to numerical thresholds defined by the selected risk threshold profile; and

- (b) enforce the selected proof-requirement policy by causing verification of the required cryptographic proofs on-chain prior to settlement, and by assigning a non-approval policy action when the required cryptographic proofs are missing or fail verification;
- (c) an offline training layer comprising a composite teacher ensemble configured to generate pseudo-labels from unlabelled blockchain transaction data using a weighted combination of at least two independent teacher models, and a student model trained on the generated pseudo-labels;
- (d) an infrastructure layer comprising a plurality of smart contracts deployed on a blockchain network, the smart contracts being configured to enforce the policy actions determined by the policy engine, including:
 - (i) a core contract configured to register compliance decisions on the blockchain;
 - (ii) an escrow contract configured to hold funds in time-locked escrow for transactions assigned the escrow policy action; and
 - (iii) a zero-knowledge proof verifier contract configured to verify cryptographic proofs on-chain;

wherein the system is configured to enforce compliance decisions *before* the corresponding transaction is settled on the blockchain network.

2. A computer-implemented method of enforcing compliance in decentralised finance transactions, the method comprising the steps of:

- (a) receiving a transaction request comprising at least a sender address, a recipient address, and a transaction value;
- (b) computing a risk score for the transaction request by applying a trained machine learning model to a feature vector comprising address-level features, transaction-level features, and graph-level features derived from the transaction request and associated blockchain data;
- (c) calibrating the computed risk score using a sigmoid recalibration function anchored at a percentile of a training score distribution;
- (d) computing one or more network instability metrics from recent transaction activity and selecting an enforcement profile from a predetermined set of enforcement profiles, the selected enforcement profile deterministically defining:
 - (i) a risk threshold profile defining numerical thresholds for a deterministic decision matrix; and
 - (ii) a proof-requirement policy defining one or more cryptographic proofs to be verified on-chain as a prerequisite to settlement for at least one risk score tier or transaction class;
- (e) mapping the calibrated risk score to a policy action selected from a predetermined set of policy actions according to the deterministic decision matrix using numerical thresholds defined by the selected risk threshold profile;
- (f) enforcing the proof-requirement policy by verifying the one or more cryptographic proofs on-chain prior to settlement, and assigning a non-approval policy action when the required cryptographic proofs are missing or fail verification;

- (g) enforcing the policy action by executing a corresponding smart contract function on a blockchain network, prior to settlement of the transaction; and
 - (h) recording the compliance decision and policy action on the blockchain as an immutable audit record.
3. A system according to claim 1, wherein the composite teacher ensemble comprises:
- (a) an autoencoder–gradient boosting teacher configured to compute anomaly scores based on reconstruction errors of an autoencoder applied to transaction features;
 - (b) a rule-based teacher configured to compute suspicion scores based on Financial Action Task Force (FATF) red-flag indicators; and
 - (c) a graph-based teacher configured to compute suspicion scores based on graph-structural properties of the transaction network;
- and wherein the composite teacher score is a weighted sum of the outputs of the at least three teacher models.
4. A system according to claim 1, further comprising a zero-knowledge proof subsystem configured to generate and verify one or more of:
- (a) a sanctions non-membership proof demonstrating that an address is not a member of a set of sanctioned addresses represented as a Merkle tree, without revealing the address;
 - (b) a risk range proof demonstrating that a risk score falls within a specified range, without revealing the exact risk score; and
 - (c) a KYC verification proof demonstrating that a participant has completed identity verification, without revealing identity data.
5. A system according to claim 4, wherein the Merkle tree of sanctioned addresses is constructed using the Poseidon hash function, and wherein the zero-knowledge proofs are Groth16 zk-SNARK proofs with a time-to-live after which the proofs must be regenerated.
6. A method according to claim 2, wherein the step of computing a risk score comprises:
- (a) computing address-level features comprising transaction count, total value transferred, mean and variance of transaction amounts, address age, number of unique counterparties, and temporal activity patterns;
 - (b) computing transaction-level features comprising transaction value, gas parameters, input data characteristics, and token transfer indicators;
 - (c) computing graph-level features by constructing a transaction graph and computing degree centrality, betweenness centrality, clustering coefficient, PageRank score, and community membership; and
 - (d) applying the trained machine learning model to the combined feature vector.
7. A system according to claim 1, wherein the deterministic decision matrix maps risk scores as follows:

- (a) scores below a first threshold to an approve action;
 - (b) scores between the first threshold and a second threshold to an approve-with-monitoring action;
 - (c) scores between the second threshold and a third threshold to a review action;
 - (d) scores between the third threshold and a fourth threshold to an escrow action; and
 - (e) scores at or above the fourth threshold to a block action.
8. A method according to claim 2, wherein verifying the one or more cryptographic proofs comprises verifying one or more zero-knowledge proofs submitted with the transaction request on-chain, wherein successful verification is a prerequisite for approving the transaction request.
 9. A system or method according to claim 1 or claim 2, wherein the one or more network instability metrics comprise at least one of: a spectral radius or centrality concentration of a transaction graph computed over a sliding time window, a temporal burst indicator computed from a transaction count process, or a shift of a calibrated risk score distribution relative to a reference distribution.
 10. A system or method according to claim 1 or claim 2, wherein selecting the enforcement profile comprises selecting one profile from a discrete set of profiles, and wherein each profile deterministically defines a distinct set of risk score thresholds for the deterministic decision matrix.
 11. A system or method according to claim 1 or claim 2, wherein the proof- requirement policy comprises, for at least one of the enforcement profiles, requiring a sanctions non-membership proof, a risk range proof, a KYC verification proof, or a combination thereof.
 12. A system or method according to claim 1 or claim 2, wherein selecting the enforcement profile further deterministically defines at least one of: a transaction rate limit, a queue delay, an escrow duration, an escrow release condition window, or a maximum transaction value limit.
 13. A system according to claim 1, further comprising a cross-chain bridge contract configured to extend compliance enforcement to cross-chain transactions by requiring compliance verification on both a source chain and a destination chain before releasing bridged funds.
 14. A method according to claim 2, wherein the trained machine learning model is a student model trained using pseudo-labels generated by a composite teacher ensemble operating on unlabelled blockchain transaction data, the composite teacher ensemble comprising at least an unsupervised anomaly detection model, a rule-based model encoding regulatory red-flag indicators, and a graph-structural model.
 15. A system or method according to any preceding claim, wherein the sigmoid recalibration function is defined as:

$$\hat{p}_{\text{cal}}(x) = \sigma(k \cdot (s(x) - P_{75}))$$

where σ is the logistic sigmoid, k is a steepness parameter, $s(x)$ is the raw model output, and P_{75} is the 75th percentile of the training score distribution.

6. DRAWINGS

[To be filed separately: System architecture diagram showing the four layers; flow diagram of the compliance decision process; schematic of the zkNAF proof generation and verification pipeline; block diagram of the composite teacher weak supervision pipeline.]

ABSTRACT

A computer-implemented system enforces regulatory compliance in decentralised finance transactions before settlement. An interface layer receives transaction requests. A compliance orchestration layer comprising a machine learning risk scoring engine, graph analytics module, sanctions screening module and state-adaptive transaction flow control module computes network instability metrics and selects an enforcement profile; the selected profile deterministically defines a risk threshold profile for the compliance decision matrix and a proof-requirement policy specifying cryptographic proofs to be verified on-chain before settlement. A deterministic policy engine maps calibrated risk scores to approve, review, escrow or block actions using thresholds defined by the selected profile. Blockchain-deployed smart contracts enforce those actions before transaction settlement, with an immutable audit trail. A zero-knowledge proof subsystem enables privacy-preserving verification of sanctions non-membership, risk score range and KYC status without revealing underlying data.